

K MOHTASHIM JAMAL

AI/ML Engineer

Bengaluru, India | +91-9916191965 | mohtashimjamal009@gmail.com | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)

PROFESSIONAL SUMMARY

AI/ML Engineer and Full-Stack Developer with hands-on experience shipping production-style, deployed applications: a full-stack GenAI research platform (Next.js, MongoDB, JWT auth, multi-agent Mastra AI orchestration) and a fine-tuned computer vision model (MobileNetV2, 97.18% validation accuracy across 28 classes). Strong Python and TypeScript fundamentals, with practical work across LLM pipelines, model fine-tuning and evaluation, and mobile development. Best Paper Award winner; fast learner with a track record of shipping and documenting real, working systems end-to-end.

TECHNICAL SKILLS

- **Languages:** Python, C++, Kotlin, TypeScript, JavaScript, HTML
- **AI, LLMs & Agents:** LLM-powered Agent Orchestration (Mastra AI Framework), Gemini API, RAG-adjacent Retrieval, Vector/Semantic Memory, Perplexity AI API, SerpAPI, Prompt Engineering
- **Machine Learning & Computer Vision:** CNNs, Transfer Learning & Fine-Tuning (MobileNetV2), Model Evaluation (Precision/Recall/F1, Error Analysis), TensorFlow, Keras, PyTorch, Scikit-learn, NumPy, TF-IDF
- **Full-Stack & Backend:** Next.js, React, Node.js, Flask, REST API Design, MongoDB, JWT Authentication, Tailwind CSS
- **Mobile Development:** Jetpack Compose, MVVM Architecture, Room, Hilt, Coroutines & Flow
- **Tools & Platforms:** Git & GitHub, Vercel, Docker (Basic), Android Studio, Cursor

PROJECTS

Intelligent Research Assistant — Best Paper Award, KNSIT Project Symposium

Project — [GitHub](#)

- Built and deployed (Vercel) a full-stack GenAI research platform (Next.js 15, React 19, TypeScript, MongoDB) with JWT authentication and persistent, user-specific research conversations.
- Designed a modular multi-agent architecture (Master, Research, Draft, Export agents via the Mastra AI framework) that searches Google Search/Scholar/News (SerpAPI, Perplexity AI fallback) and synthesizes findings into citation-backed reports (Gemini) exportable to PDF/HTML/Markdown.
- Collaborated with a 4-person team; won Best Paper Award at the KNSIT Project Symposium.

Fruit & Vegetable Quality Classifier — Personal Project

Project — [GitHub](#)

- Fine-tuned MobileNetV2 (TensorFlow/Keras) on a custom 29,291-image dataset to classify 28 classes (14 fruit/vegetable types × fresh/rotten): froze the pre-trained backbone for initial transfer learning, then unfroze the last 30 layers at a low learning rate (1e-5) for fine-tuning.
- Achieved 97.18% validation accuracy and 95.58% macro F1-score on 5,859 held-out images; built a full evaluation pipeline (Scikit-learn) reporting per-class precision/recall/F1 and analyzing common misclassifications.
- Built a full-stack Flask web app for image upload, inference, confidence scoring, and nutrition-info lookup.

EXPERIENCE

Android Developer Intern — MindMatrix (VTU-Affiliated Virtual Internship Program) Feb 2026 – May 2026

Project: *KreedaPrerana* — AI-Assisted Sports Talent Scouting App — [GitHub](#)

- Architected a native Android app (Kotlin, Jetpack Compose, MVVM) for sports talent scouting, with an offline-first data layer (Room, Hilt) and a rule-based evaluation engine benchmarking live results against performance datasets; rated "Excellent" by internship coordinators.
- Built a dynamic leaderboard and performance-trend visualizations (MPAndroidChart, Kotlin Flow), architecting an AI-insights module for pluggable LLM integration (Gemini).

EDUCATION

B.E., Artificial Intelligence & Machine Learning — KNS Institute of Technology, Bengaluru 2022 – 2026

- CGPA: 8.34/10.0
- Relevant Coursework: Data Structures & Algorithms, Deep Learning, Natural Language Processing, Database Management Systems

CERTIFICATIONS

- Introduction to Machine Learning (NPTEL, IIT Madras) & Cloud Computing (NPTEL, IIT Kharagpur)
- AI: Search Methods for Problem Solving, Data Analytics with Python & GPU Architectures — VTU Centre for Online Education